Enhancing GNSS Receiver Resilience through Fuzz Testing: A Novel Approach to System Robustness in Avionics

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***Abstract*—Global Navigation Satellite System (GNSS) receivers are essential to modern aviation, yet remain vulnerable to spoofing, jamming, and other interference. Traditional validation methods often miss complex or implementation-specific failure modes. This paper presents a sensitivity-informed framework to support fuzz testing of GNSS receivers, focusing on the Satellite-Based Augmentation System (SBAS). A one-factor-at-a-time (OFAT) sensitivity analysis was conducted to identify SBAS message parameters with the greatest impact on key receiver outputs, including protection levels, position accuracy, clock bias, and navigation continuity. These findings informed the prioritization and reduction of fuzzing input cases. Although only structured, mutation-based black-box fuzzing was implemented, the results demonstrate its feasibility and effectiveness in uncovering latent vulnerabilities. The approach provides a scalable foundation for future robustness testing in safety-critical GNSS applications.**

***Index Terms*—GNSS, SBAS, Fuzz Testing, Receiver Robustness, Sensitivity Analysis, Integrity Monitoring, GNSS Validation, Aviation Safety**

I. Introduction

Global Navigation Satellite System (GNSS) receivers are critical to aviation, providing essential positioning, navigation, and timing (PNT) services that support safety and operational efficiency. However, their reliability is increasingly compromised by threats such as jamming and spoofing, where attackers either block signals or transmit counterfeit ones to mislead receivers [1]. Recent events underscore the severity of these risks: thousands of flights over Eastern Europe have reported GNSS disruptions suspected to result from deliberate interference [2], while the European Union Aviation Safety Agency (EASA) Safety Information Bulletin 2022-02R2 documents rising interference incidents across Eastern Europe and the Middle East [3]. Commercial operators have similarly reported widespread spoofing in active airspace [4]. Even brief outages in these environments can erode situational awareness and compromise system integrity, highlighting the urgent need for robust testing methodologies to ensure GNSS receiver resilience.

Despite the importance of GNSS receiver validation, standard methods like conformance testing and performance benchmarking often miss subtle failure modes. Their focus on deterministic scenarios limits their ability to detect issues caused by rare, malformed, or edge-case inputs;,leaving latent vulnerabilities undetected during development and certification.

To address this gap, this work explores fuzz testing as a complementary evaluation tool for GNSS receivers. Fuzzing—widely used in software security—involves injecting randomized, malformed, or adversarial inputs to uncover unexpected behavior. This aggressive testing not only verifies system responses within specification limits, but also reveals edge cases and unforeseen issues beyond traditional test coverage. It has proven effective in detecting vulnerabilities such as buffer overflows, denial of service (DoS), and code injection.

In GNSS systems, fuzzing can uncover hidden flaws by simulating atypical or hostile signal conditions that deterministic testing may miss. However, applying it poses challenges due to the receivers' hybrid, time-dependent nature and continuous signal processing, requiring substantial computational resources. Additionally, the opaque architecture of commercial receivers limits white-box or grey-box approaches. Effective fuzzing therefore relies on black-box strategies guided by domain knowledge and targeted parameter selection.

In this context, we aim to study a technique to reduce the number of tests by focusing on the parameters that have the greatest impact. Concretely, our work focuses on identifying high-impact SBAS fields to guide GNSS fuzz testing. A one-

factor-at-a-time (OFAT) sensitivity analysis identifies critical parameters whose perturbation leads to meaningful deviations. These insights inform the design of structured, black-box, mutation-based fuzzing.

The complexity of GNSS systems, which process signals from multiple satellites under varying environmental conditions, presents unique challenges for fuzz testing. Effective application requires identifying critical input parameters susceptible to anomalies. Sensitivity analyses can pinpoint these parameters, optimizing the fuzz testing process by focusing on the most vulnerable aspects of the receiver's operation. This targeted approach ensures efficient testing without compromising coverage.

The findings confirm that even a targeted fuzzing approach, when guided by sensitivity results, can surface vulnerabilities beyond the reach of standard validation.

II. State of the Art

Traditional testing methods, including conformance and performance evaluations, often fall short in identifying complex vulnerabilities that sophisticated attacks exploit. To address this gap, fuzz testing has emerged as a potent technique in software security, involving the injection of malformed or unexpected inputs to uncover hidden flaws [5].

Though widely used in software assurance, fuzz testing remains underexplored for complex systems like GNSS receivers. Black-box, grey-box, and white-box techniques offer varying benefits: black-box mimics external attacks, white-box uses full system knowledge, and grey-box blends partial insight with feedback loops [6]. Assessing these methods in GNSS contexts is key to uncovering vulnerabilities.

Recent efforts, including novel techniques presented at ION ITM 2025 [7], begin addressing this gap. Integrating fuzz testing into GNSS evaluation frameworks may systematically reveal weaknesses by simulating unpredictable signal scenarios, thereby improving robustness.

Sensitivity analysis in fuzzing refers to techniques that measure how program behavior changes in response to small perturbations in input data. The goal is to identify inputs that have a significant impact on program execution — e.g., control flow changes, memory access, or value propagation — and use this to guide test generation strategies. This technique is very similar to Taint analysis [8], which tracks how input data influences the program's internal variables and its execution. The difference lies in the fact that sensitivity analysis involves the notion of the importance of this influence, seeking to identify which changes in the input have the greatest influence.

Sensitivity analysis based tools analyze execution traces to correlate input data with changes in execution paths. For example *Angora* [9] uses byte-level taint tracking and gradient descent to pinpoint "sensitive" input data. In *NEUZZ* [10], small changes to inputs are treated like "perturbations," and their effect on the execution is measured via cost functions (e.g., branch distance) with a neural network.

Recent work [11] introduces a novel approach to context-sensitive fuzzing by leveraging function cloning to enable precise, collision-free context-sensitive coverage tracking. The authors propose a prediction scheme that identifies promising contexts for fuzzing, addressing the challenges of state explosion and improving fuzzing effectiveness, as demonstrated by their approach outperforming state-of-the-art fuzzers and uncovering several security vulnerabilities.

In numerical domains, *Numerical Fuzz* [12] introduced a type-based approach to capture how small changes in input influence outputs—an idea particularly relevant for GNSS applications.

Type systems, such as the *Hindley-Milner sensitivity system* [13], help identify structurally sensitive inputs, enabling test set reduction without compromising variability. In [14], the authors extend the Fuzz type system by incorporating L^p distances for sensitivity analysis, enhancing its ability to handle vector metrics and improving the accuracy of privacy and optimization analyses in probabilistic programs.

Finally, coverage-guided fuzzers, like *AFL* [15], adapt based on runtime behavior. Building on these techniques, model-driven sensitivity further refines input selection, making it particularly effective for the nuanced and highly variable environment of GNSS fuzzing.

Although fuzz testing for GNSS receivers remains largely unexplored, it holds significant potential to improve aviation safety by uncovering hidden vulnerabilities. It complements traditional methods by exposing system weaknesses under adversarial conditions, and sensitivity analysis strengthens this approach by identifying high-impact inputs to guide effective test generation.

III. Methodology

This section describes the experimental framework used to evaluate the sensitivity of GNSS receiver performance to individual SBAS message fields. The goal of this analysis is to systematically identify which SBAS parameters exert the greatest influence on key performance metrics, and to support the reduction and prioritization of test cases—particularly in the context of fuzz testing for receiver robustness.

A. One-Factor-at-a-Time Sensitivity Approach

The sensitivity study was conducted using a one-factor-at-a-time (OFAT) method. In this approach, a single SBAS message parameter is modified while all other parameters remain fixed to a reference configuration. This allows for the isolated observation of each field's influence on receiver outputs, without interaction effects.

Thirteen distinct SBAS message parameters were selected, based on their functional relevance and known structure as outlined in ESA SBAS documentation [16], including:

- MT2 – Fast Corrections (FC_F1)
- MT5 – Fast Corrections (Residual)
- MT6 – User Differential Range Error Indicator (UDREI)
- MT6 – Issue of Data (IODF)
- MT9 – User Range Accuracy (URA)
- MT9 – GEO Satellite Parameters (GeoSatParam_X)
- MT9 – GEO Satellite Accuracy (GeoSatAcc_X)

- MT9 – Clock Parameters
- MT10 – Range Correction Timing (RTC)
- MT12 – Common Parameters (e.g., IGP mask, a0 term)
- MT24 – Additional Fast Correction Parameters (FC_F1)
- MT25 – Reduced Orbit Parameters: Delta-Z
- MT25 – Reduced Orbit Parameters: Velocity Code

Each parameter was perturbed from the reference case to introduce a change representative of a degraded or altered operational condition. The perturbation magnitudes were chosen based on plausible deviations that might occur due to signal anomalies, operational errors, or intentional fuzzing inputs.

B. Test Setup and Evaluation Metrics

The test environment consisted of a GNSS receiver setup capable of processing SBAS corrections. The receiver was provided with SBAS messages containing controlled field modifications. Each test run produced time series data for multiple output metrics, which were then analyzed and compared to the unmodified reference scenario.

The following GNSS performance indicators were evaluated for sensitivity:

- **Protection Levels (VPL/HPL)**: Vertical and horizontal bounds on positioning uncertainty, computed as part of the receiver's integrity monitoring function.
- **PDOP**: Position Dilution of Precision, a measure of satellite geometry and its effect on horizontal positioning accuracy.
- **Navigation State**: A discrete indicator of whether the receiver is able to compute a valid navigation solution (state 3).
- **Clock Bias**: The estimated time offset between the receiver's internal clock and GNSS system time.
- **AGC Level 1**: The receiver's internal automatic gain control, reflecting signal amplitude and dynamic range behavior.
- **Positioning Accuracy**: Deviations in Earth-Centered Earth-Fixed (ECEF) and Latitude/Longitude/Height (LLH) coordinates.

To enable systematic comparison across SBAS parameters and output metrics in the context of GNSS fuzz testing, each result was assigned a qualitative sensitivity score from 0 (no effect) to 10 (very high effect). Scores were based on the observed magnitude, consistency, and prominence of output changes during one-factor-at-a-time testing. Though interpretive, the process was applied consistently to reflect relative impact. The scale used was:

- **9–10** = very high (pronounced effects),
- **7–8** = strong (clear but not dominant effects),
- **5–6** = moderate (measurable but less consistent changes),
- **3–4** = low (minor or context-dependent effects),
- **0–2** = negligible (little to no observable impact).

This structured approach supports comparative analysis of fuzzing sensitivity and highlights parameters likely to affect system robustness. A quantitative evaluation of these results is planned for future work.

IV. Fuzzing Strategy and Feasibility

While fuzz testing offers significant potential for uncovering hidden vulnerabilities in GNSS receivers, not all fuzzing strategies are equally feasible in practice—especially when applied to complex, safety-critical, and largely opaque systems like aviation-grade GNSS receivers. This section discusses the practical limitations observed during the adaptation of fuzzing techniques and justifies the choice of methods used in this study.

GNSS receivers, particularly those designed for aviation, are highly optimized and rigorously validated systems. Their robustness means that naive fuzzing—random bit-level mutation or fully unguided input generation—rarely yields meaningful test results. To generate impactful perturbations, fuzzing must be informed by domain knowledge such as protocol structure, signal timing, and navigation model behavior.

In theory, feedback-driven techniques such as Coverage-Guided Fuzzing (CGF) or reinforcement learning could be highly effective. These approaches adapt based on internal receiver states or crash indicators to explore deeper execution paths. However, GNSS receivers are typically treated as black boxes in test environments, with no access to internal memory, software coverage metrics, or processing logic. Integrating a real-time feedback loop requires significant instrumentation, which is very exhaustive to implement.

Instead, this study adopts a purely **black-box fuzzing approach**, treating the GNSS receiver as an isolated system under test. A baseline scenario was defined using known-good SBAS messages and satellite geometries. Each test case was then derived by modifying a single SBAS field based on a predefined distribution model. These modifications simulate degraded or adversarial conditions while preserving protocol validity.

The fuzzed scenarios were generated using an in-house fuzzing tool. This tool allows precise control over which SBAS field to fuzz, the probability distribution applied (e.g., uniform, Gaussian), and the number of generated test variants. This semi-structured approach strikes a balance between input diversity and operational realism, ensuring the generated cases remain syntactically valid while still stressing the receiver in meaningful ways.

The fuzz testing framework used in this study can be classified as **structured, mutation-based probabilistic black-box fuzzing**. Test inputs were generated by selectively mutating specific SBAS message fields using domain-informed probability distributions, while ensuring protocol compliance. This strategy preserves a black-box interaction with the GNSS receiver and enables systematic exploration of high-impact perturbations without relying on internal feedback or instrumentation. By applying probabilistic models to drive the mutation process, the approach allows for the generation of diverse and randomized test cases that efficiently probe the input space. In the conducted experiments, this combination proved to be both feasible and effective, producing observable

deviations in critical output metrics when perturbing sensitive parameters such as URA and UDREI.

In summary, while advanced fuzzing techniques may hold theoretical promise, black-box and probabilistic fuzzing currently offer the most practical and scalable approach for GNSS receiver testing. This aligns with real-world test setups where the receiver behaves as a sealed unit, and test inputs must be delivered via well-defined signal simulations or protocol injections without relying on internal telemetry.

V. Test Setup

The test campaign was conducted using a hardware-in-the-loop simulation environment tailored for evaluating SBAS-specific GNSS receiver behavior under controlled and repeatable conditions. The goal was to assess how variations in individual SBAS message fields influence receiver performance across critical operational outputs.

A. GNSS Receiver Under Test

The system under test is a dual-frequency, multi-constellation GNSS receiver compliant with emerging aviation standards, including EUROCAE ED-259. The device supports SBAS signal processing on L1 and L5 frequencies and inherits safety-critical features from legacy DO-229F-compliant designs.

B. Test Bench and Signal Simulation

All tests were performed using a Spirent GSS-9000 GNSS signal simulator with PosApp software (v10). The simulator was configured to generate realistic satellite signal environments, including GPS constellation and SBAS augmentation signals. Only SBAS messages were varied during the test runs, while GPS navigation and satellite geometries remained constant.

The simulator's "Data Modification" interface was used to selectively alter SBAS message content during runtime. This feature enables fine-grained, field-level changes in specific message types without affecting overall message structure or protocol compliance. For each SBAS message field, a reference scenario was defined using nominal values. Each test run then introduced a controlled variation in exactly one field, keeping all other fields fixed.

C. Fuzzing Tool and Input Generation

To support field-level SBAS fuzzing, a custom-developed internal fuzzing tool purpose-built for GNSS testing was employed.

This tool provides structured and probabilistic fuzzing capabilities designed for black-box testing. Its key features include:

- Selection of a single SBAS message field as the fuzz target.
- Assignment of a probability distribution (e.g., normal, uniform) to generate realistic or adversarial deviations.
- Controlled test case generation using sampled values from the distribution.

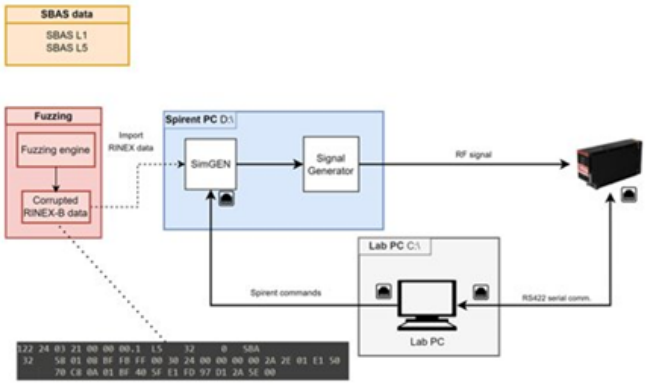


Fig. 1: Hardware-in-the-loop test setup used for SBAS fuzzing.

This approach ensured that test inputs remained syntactically correct but contained controlled, targeted anomalies. Importantly, the tool was integrated with the Spirent simulator's modification interface.

D. Execution and Observables

During each simulation, the receiver processed the SBAS-corrected GNSS signals and produced output logs capturing time series data for the following observables:

- Protection Levels (VPL, HPL)
- Position Dilution of Precision (PDOP)
- Clock Bias
- Automatic Gain Control (AGC) Level 1
- Navigation State (e.g., fix/no-fix, state transitions)
- Positioning error in ECEF and LLH coordinates

Truth logs generated by the Spirent simulator ensured that deviations from expected behavior could be quantified for each fuzzed input. All test cases were compared against the baseline reference scenario to determine field-specific sensitivity.

E. Testing Scope and Goals

The testing scope was intentionally limited to SBAS message manipulation, allowing for a focused evaluation of how each augmentation field influences core GNSS receiver metrics. No GPS or Galileo navigation message content was altered. This decision was made to isolate the role of SBAS content in influencing receiver integrity, timing, and position estimation behavior.

The overarching objective was to use this targeted fuzzing campaign to identify which SBAS fields most critically affect receiver behavior—thereby informing test prioritization, guiding robustness validation, and enabling more efficient fuzzing campaigns in future GNSS certification and development workflows.

VI. Results and Analysis

This section presents the results of a one-factor-at-a-time (OFAT) sensitivity analysis conducted across multiple SBAS message parameters. A consolidated sensitivity matrix is provided in Table I, summarizing the qualitative impact scores (0

= no effect, 10 = very high effect) for each SBAS field across all outputs.

A. VPL and HPL

The strongest influence on VPL and HPL was observed when modifying the User Range Accuracy (URA) parameter contained in SBAS Message Type 9 (MT9), and the UDRE Indicator found in Message Type 6 (MT6). These fields directly contribute to the modeling of pseudorange error and signal trust, and their perturbation led to a significant increase in protection levels — in some cases, by an order of magnitude. This underlines their critical role in determining integrity bounds and highlights the importance of precise calibration and monitoring, particularly in safety-of-life applications.

Moderate sensitivity was observed when altering the Fast Correction parameters in MT2 (MT2_FC_F1) and the Range Correction Timing in MT10 (MT10_RTC). These components influence the timing and ephemeris correction process and become more impactful under degraded signal or sparse update conditions.

Conversely, adjustments to fields in MT25 (e.g., reduced orbit parameters), Clock Parameters, and MT9 Geo Satellite position data (GeoSatParam_X) produced negligible deviation from the reference case. These parameters exhibited low sensitivity in the tested configuration and appear relatively robust under nominal system behavior.

This analysis confirms that specific SBAS message content can substantially affect the integrity margins calculated by GNSS receivers. In particular, the URA and UDRE fields — found in MT9 and MT6 — should be prioritized for optimization and continuous quality assurance in safety-critical operations. Future work will include statistical generalization of these impacts and potential interaction effects using factorial or global sensitivity analysis frameworks.

B. PDOP

To evaluate the impact of SBAS message perturbations on the Position Dilution of Precision (PDOP), it was analyzed how modifications to individual message fields influenced the receiver's perception of satellite geometry. The PDOP metric is particularly sensitive to satellite positioning and weight assignment in the navigation solution, making it a useful indicator of geometric distortion or reliability loss.

The reanalysis uncovered several patterns. In particular, Message Type 12 (Common Parameters) and Message Type 24 (Fast Correction F1) demonstrated substantially different scaling behavior compared to both the baseline and other message types. These configurations consistently produced elevated PDOP values throughout the simulation, indicating that the satellite geometry model used by the receiver was significantly perturbed by these parameter changes.

In contrast, MT6 UDREI and MT9 Clock Parameters resulted in PDOP curves that were nearly flat and closely aligned with the reference scenario. This suggests these fields had minimal influence on the satellite weighting factors affecting PDOP under the test conditions.

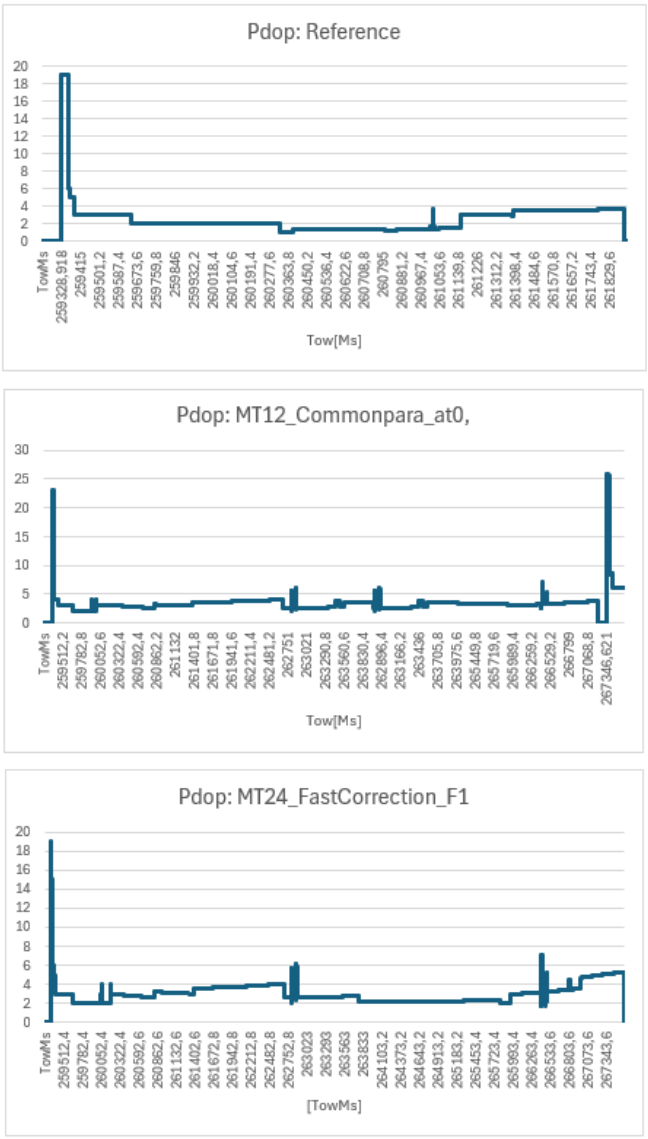


Fig. 2: PDOP sensitivity comparison: MT12_Common_a0, MT24_FC_F1, and Reference. Elevated PDOP under MT12 and MT24 indicates significant geometric distortion compared to the baseline.

Other message types, including MT2 Fast Corrections, MT10 RTC, and MT25 fields, also demonstrated noticeable but more moderate deviation from the baseline. Their trajectories were distinguishable from the reference but did not show the extreme scaling observed in modifications of MT12 and MT24.

These findings underscore that while UDRE and clock parameters play a role in signal integrity, their influence on satellite geometry metrics like PDOP is limited. Instead, structural and timing-related fields, such as those in MT12 and MT24, can cause substantial distortions in geometric evaluation—highlighting them as critical fields for further robustness testing.

TABLE I: Consolidated Sensitivity Scores of SBAS Parameters Across Output Metrics (0 = No Effect, 10 = Very High Effect)

SBAS Parameter	VPL/HPL	PDOP	Nav	ClkBias	AGC	ECEF X	ECEF Y	ECEF Z	LLH X	LLH Y	LLH Z
MT9_URA	10	5	8	5	5	5	9	9	9	9	3
MT6_UDREI	9	2	7	4	9	3	8	8	3	3	5
MT2_FC_F1	6	6	6	4	5	4	6	6	4	4	5
MT10_RTC	7	5	9	9	8	6	6	5	6	6	7
MT24_FC_F1	3	9	6	3	9	3	4	4	3	3	4
MT25_rOrbit_DeltaZ	2	5	7	2	4	3	3	3	2	2	10
MT9_ClockParam	3	2	5	9	6	8	7	4	8	8	8
MT12_Common_a0	4	10	3	2	3	8	6	3	2	2	6
MT5_FC_F1	3	4	3	3	4	3	4	4	2	2	3
MT6_IODF	3	4	3	3	10	3	4	6	4	4	3
MT9_GeoSatParam_X	2	3	2	7	5	2	4	3	8	8	2
MT9_GeoSatAcc_X	3	2	4	8	6	8	9	4	8	8	8
MT25_rOrbit_VC	3	5	3	2	4	2	3	3	2	2	9
MT5_FastCorr_Residual	2	4	2	2	3	2	2	2	2	2	9

C. Navigation State

Among all SBAS parameters tested, the most severe and persistent degradation in navigation state was observed in scenarios where the Reduced Orbit Parameter Delta-Z in Message Type 25 (MT25) and the Range Correction Timing (RTC) in Message Type 10 (MT10) were perturbed. In particular, MT25 exhibited the strongest effect, causing sustained periods during which the receiver was unable to compute a valid navigation solution (state 3). These disruptions were not transient but persisted throughout extended simulation intervals, indicating a structural failure in the receiver's ability to reconcile satellite geometry under these conditions.

Similarly, alterations to the Range Correction Timing (RTC) field in MT10 produced prolonged outages in the navigation state. The impact was more fragmented than in the MT24 case but still demonstrated high sensitivity, highlighting the importance of accurate timing alignment in the SBAS correction chain. In contrast, in Message Type 6 (MT6)—the User Differential Range Error Indicator (UDREI)—and Message Type 2 (MT2)—containing fast corrections—also caused deviations from the reference case, but the effects were more intermittent. The receiver often managed to reacquire a valid navigation state after short outages, suggesting that while these fields are critical for integrity assessment, their perturbation does not inherently cause structural navigation failure unless combined with other degradations.

Lastly, perturbations in the Geo Satellite Accuracy X-component from MT9 produced milder impacts. While they contributed to variations in the stability of the navigation solution, they did not result in complete state loss. This suggests their influence is more related to satellite confidence and positional weighting rather than core navigation continuity.

These results highlight that MT25 Reduced Orbit Parameter Delta-Z and MT10 RTC represent the most sensitive fields with respect to navigation state availability, followed by MT6 and MT2, while MT9 (GeoSatAcc X) had the least disruptive effect in this context.

D. Clock Bias

Clock bias—the estimated time offset between the receiver and GNSS system time—was also analyzed. For each test, one field was modified while the rest of the SBAS content remained unchanged.

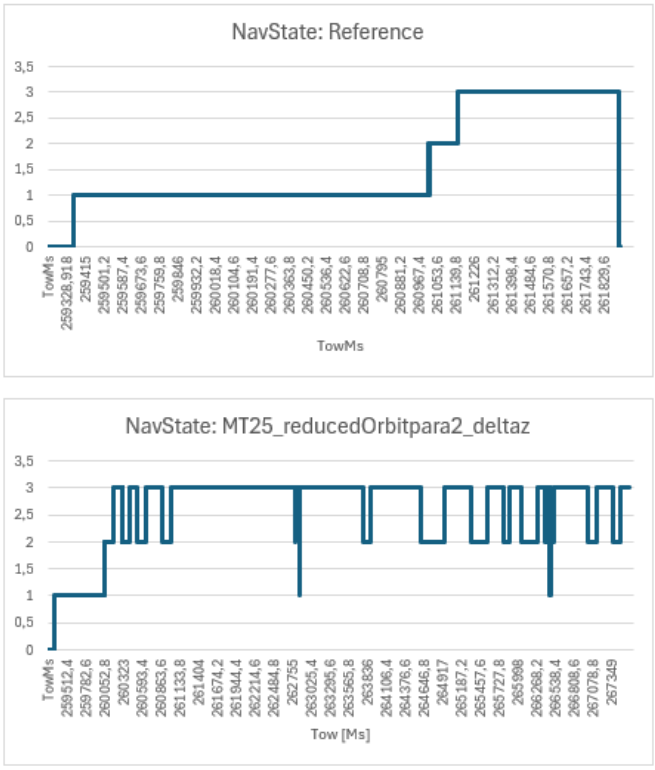


Fig. 3: Navigation state availability under perturbation of MT25_rOrbit_DeltaZ compared to the reference case. MT25 causes extended loss of valid navigation solutions (state 3).

Contrary to expectations, parameters previously assumed to have limited influence—such as MT9 Geo Satellite Accuracy X (GeoSatAcc_X) and Clock Parameters (Clockparam)—exhibited some of the most significant deviations in clock bias, with observable double peaks in the time series. Likewise, MT10 RTC showed time-localized distortions, reinforcing its role in timing sensitivity. These findings suggest that under certain perturbations, parameters beyond the standard URA and UDREI fields can notably influence

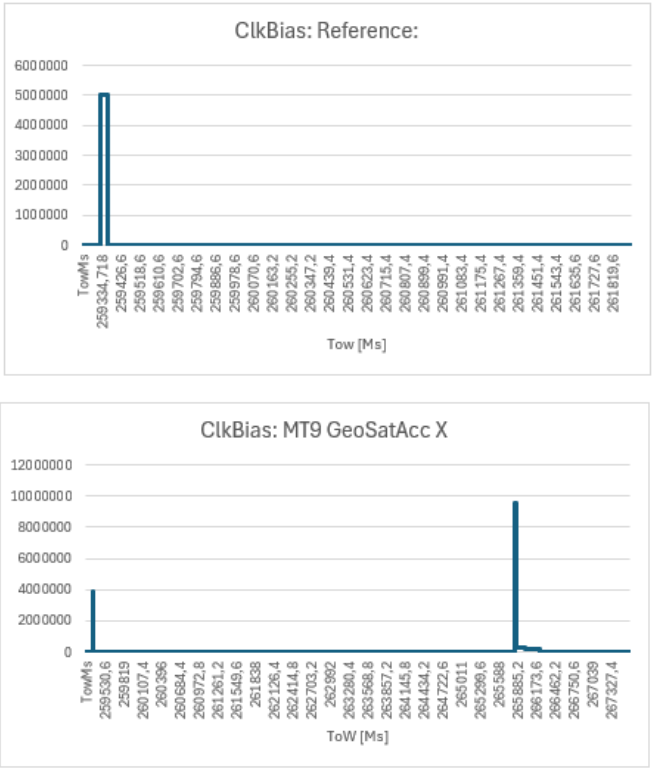


Fig. 4: Clock bias behavior under perturbation of MT9_GeoSatAcc_X compared to the reference case. MT9_GeoSatAcc_X introduces double-peaked deviations, indicating significant timing sensitivity.

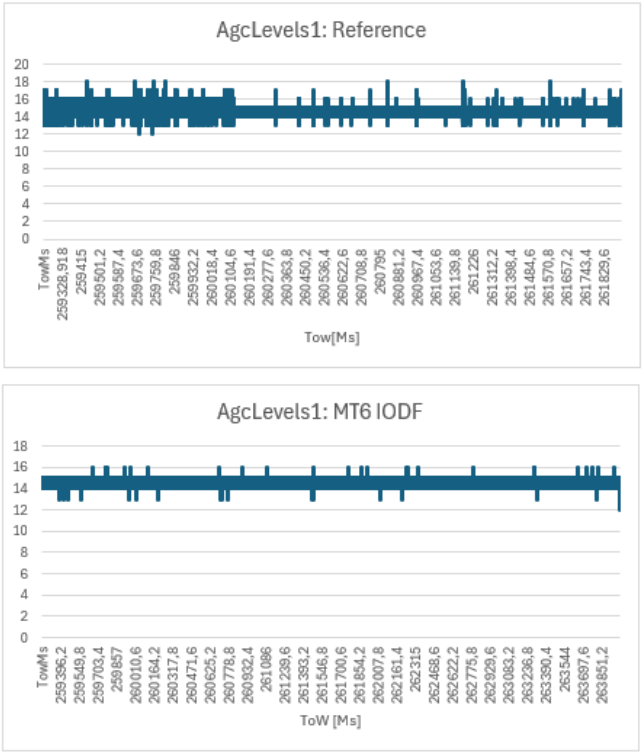


Fig. 5: AGC Level 1 response under perturbation of MT6_IODF compared to the reference scenario. MT6_IODF induces strong and consistent deviations, indicating high sensitivity in signal gain adaptation mechanisms.

GNSS receiver timing behavior, warranting their inclusion in robustness assessments.

1) Automatic Gain Control: AGC Level 1, which reflects internal signal gain adjustments, was evaluated to determine the receiver's response to signal strength and quality variations caused by SBAS content changes.

The AGC Level 1 sensitivity analysis revealed that the MT6_IODF field had the most consistent and strong deviation from the reference scenario, indicating that manipulating integrity indicators can significantly affect signal gain behavior. Additionally, MT24_FC_F1 and MT2_FC_F1 contributed visible, though less consistent, disturbances. In contrast, fields like MT9_GeoSatParax and MT9_URA had almost negligible impact, aligning closely with the reference across the test duration.

E. ECEF Position X & Z

The ECEF X- and Z-axis results exhibited similar sensitivity profiles, allowing them to be interpreted jointly due to the near-identical trajectory of deviations across test cases. The most pronounced divergence from the reference was observed for MT12 (common parameters), MT9 Geo Satellite Accuracy X (GeoSatAcc X), and MT9 clock parameters. These fields produced a characteristic dip toward the end of the simulation, indicating their influence on long-term spatial accuracy.

MT10 (Range Correction Timing) also caused moderate positional shifts, especially in ECEF Z, where a more noticeable drift pattern appeared. While not as severe as the top three contributors, the impact of MT10 confirms its role in timing consistency and positional integrity over time.

In contrast, other parameters caused only minor or negligible changes. The alignment between ECEF X and Z responses supports their grouped interpretation and points to shared underlying mechanisms—most likely linked to ephemeris timing and geometric modeling consistency.

F. ECEF Position Y

The sensitivity analysis of the ECEF Y-coordinate reveals several parameters with deviations from the reference profile. Most prominently, MT9 GeoSatAcc X exhibits the strongest divergence, showing a sustained offset pattern toward the end of the time series. This suggests a structural effect on the satellite position model impacting east-west positioning components.

Additionally, MT9 Clock Parameters, MT12 Common Parameters, and MT10 RTC also present significant deviations. The MT9 Clock Parameters introduces timing shifts that propagate into positioning errors, particularly along the Y-axis. MT12 and MT10—while exhibiting different patterns than the reference—show elevated variability and periodic oscillations,

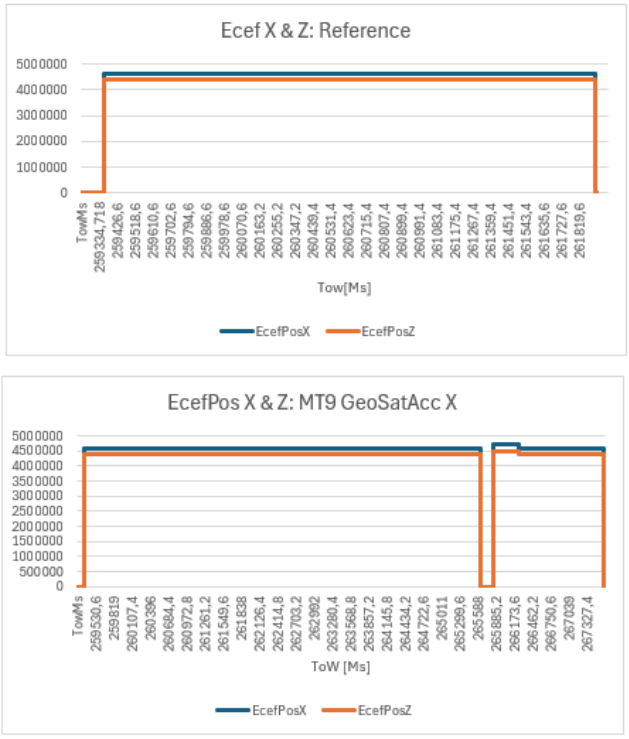


Fig. 6: ECEF X and Z positioning errors under perturbation of MT9_GeoSatAcc_X compared to the reference case. Both axes exhibit a characteristic deviation pattern, indicating long-term spatial distortion driven by satellite accuracy perturbations.

suggesting perturbations in the estimation of satellite geometry or correction application timing.

Although less impactful than in the X and Z components, MT2 Fast Correction and MT6 UDREI also contribute moderate deviations. Their effects are consistent with changes in weighting and confidence of the correction data but are less pronounced in the lateral dimension.

Given the similar behavior across ECEF X, Y, and Z for several parameters, it's evident that clock timing, correction accuracy, and satellite ephemeris perturbations jointly impact spatial dimensions. As such, ECEF X and Y are closely coupled in their response to certain SBAS modifications—particularly those involving MT9 elements—supporting their treatment as a coupled group in sensitivity evaluation.

G. LLH X (Longitude)

The LLH X-coordinate, which corresponds to longitudinal positioning, exhibited deviations in response to specific SBAS parameter modifications. In particular, the MT9 Geo Satellite Accuracy (GeoSatAcc X) parameter caused a clear deviation pattern, characterized by an early deviation followed by a persistent dip, indicating a strong influence on the longitudinal solution. Similarly, MT9 Clock Parameters and MT12 Common Parameters (a0 term) also introduced a marked and consistent deviation with a recognizable downward plateau, suggesting systematic impacts on the longitude estimate.

Moreover, MT10 RTC (Range Correction Timing) presented an abnormal fluctuation with a distinguishable dip, reinforcing its role in temporal precision which indirectly affects positional calculation, especially in longitudinal displacement under this configuration.

These perturbations are structurally different from the reference scenario and from one another but show a consistent trend of affecting the receiver's east-west positioning, confirming their significance. Consequently, these SBAS parameters should be treated as high-sensitivity factors when assessing longitudinal robustness in GNSS receivers.

H. LLH Y (Latitude)

The LLH Y-coordinate, which represents latitude in the geodetic coordinate frame, showed a very similar sensitivity profile to LLH X, indicating that the horizontal position solution is impacted in a consistent manner across both axes. Most notably, the MT9_GeoSatAccX parameter exhibited a significant dip toward the end of the simulation period, reflecting strong deviation from the reference case. This behavior mirrors the influence observed in LLH X and confirms the dominant role of satellite geometry information in affecting lateral positional stability.

Furthermore, MT12_Commonpara_a0 and MT9_Clock Parameters also showed structured divergence from the reference, forming either distinct dips or prolonged plateaus. These parameters influence the satellite selection and correction model applied by the receiver, thereby subtly shifting the computed position over time. MT10_RTC had a more moderate impact, introducing minor but visible deviations throughout the run.

Given the close similarity in behavior between LLH X and LLH Y outputs, both can be jointly considered in the evaluation of horizontal positional integrity. The consistently high impact of MT9-related parameters suggests they are crucial to receiver robustness in lateral positioning.

I. LLH Z (Height)

The vertical positioning component (LLH Z) exhibited considerable sensitivity to multiple SBAS message parameters, with several message types causing large-scale deviations from the reference. The most pronounced influence was observed in MT25 Reduced Orbit Parameters, particularly the velocity and code subfields. These exaggerated deviations reflect the known challenge of resolving vertical position in GNSS, where satellite geometry provides weaker observability in the vertical axis compared to horizontal dimensions. This suggests that inaccuracies in orbit velocity and satellite identification data can propagate substantially into vertical position estimates. Particularly strong deviations in LLH Z call for focused testing of MT25 subfields in aviation landing scenarios. Additionally, the Clock Parameters in MT9 and the Range Correction

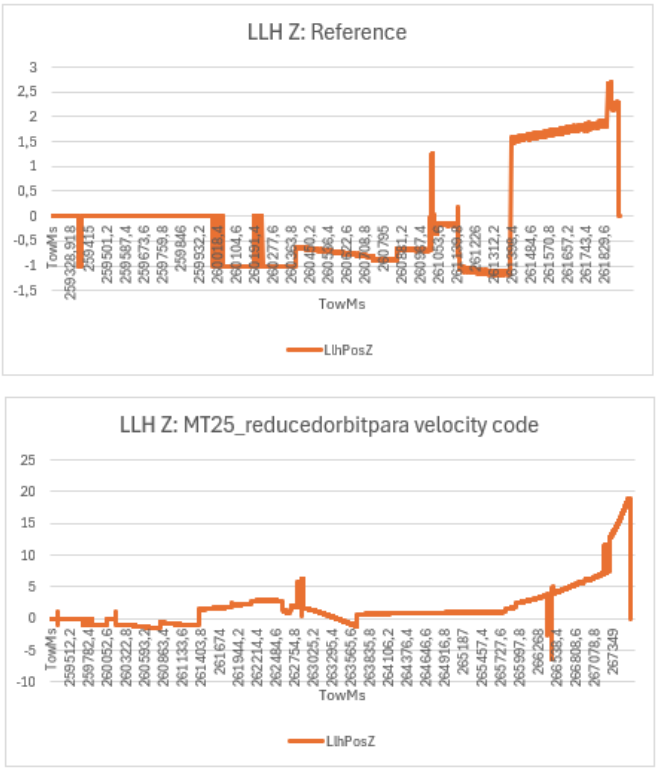


Fig. 7: LLH Z (Height) deviations under perturbation of MT25_rOrbit_VC compared to the reference case. The MT25_rOrbit_VC field causes significant vertical position drift, emphasizing the sensitivity of height estimation to orbital velocity and code errors.

Timing in MT10 (RTC) also led to persistent vertical drifts. These trends likely stem from distortions in satellite clock modeling and timing synchronization, which are critical for resolving height, given its higher sensitivity to timing errors compared to horizontal dimensions.

Further deviation was caused by MT12 Common Parameters, which introduced noticeable divergence and plateau-like behavior. The observed trends suggest that broadcast configuration values—particularly those influencing ionospheric grid mask and interpolation—have non-negligible effects on altitude resolution.

While moderate effects were seen in MT2 Fast Corrections and MT6 UDREI, these variations were less extreme but still noteworthy, indicating secondary sensitivity of vertical accuracy to fast-changing correction messages and user differential error modeling.

In contrast, fields such as MT5 and M6 IODF exhibited only low to moderate impact on LLH Z, while preserving general alignment with the reference trajectory.

Overall, the LLH Z domain proves highly susceptible to perturbations in orbit dynamics, clock modeling, and configuration timing, reaffirming the need for precise calibration and monitoring of these SBAS components in safety-critical applications like aviation navigation.

VII. Discussion

The updated sensitivity matrix and results indicate that certain SBAS message parameters have a consistent and measurable impact across multiple GNSS receiver performance outputs, while others show more metric-specific or limited effects.

Across most categories, the parameters with the highest multi-domain impact include **MT9_URA**, **MT6_UDREI**, **MT10_RTC**, **MT24_FC_F1**, and **MT9_ClockParam**. These fields consistently caused strong deviations, particularly in navigation availability, positioning accuracy (both ECEF and LLH), and timing stability (Clock Bias).

Notably, **MT10_RTC** emerged as a more influential factor than previously suggested in some domains—most significantly in navigation continuity, clock bias estimation, and vertical accuracy. This was corroborated by visual indicators such as persistent drifts and time-localized spikes.

MT12_Common_a0 displayed pronounced influence on PDOP and ECEF positioning dimensions, suggesting its critical role in satellite geometry modeling and correction consistency. Meanwhile, the MT25 fields—especially the velocity and code subfields—dominated the vertical (LLH Z) sensitivity profile, further reinforcing their importance in height determination under SBAS augmentation.

Parameters such as **MT9_GeoSatParam_X** and **MT5_FastCorr_Residual** were less impactful overall, although they did introduce specific perturbations in LLH horizontal positioning, warranting limited attention in certain robustness tests.

Key takeaways from the reevaluated results include:

- **Horizontal coupling:** ECEF X, Y and LLH X, Y showed tightly correlated sensitivities—especially under MT9 and MT10-related perturbations—suggesting common propagation pathways linked to clock modeling and correction delay.
- **MT10 RTC and MT9 ClockParam:** Both are now more clearly recognized as critical across multiple dimensions, not only in time alignment but also spatial integrity.
- **AGC sensitivity:** The MT6_IODF field remains a strong outlier in AGC behavior, revealing how even control indicators can affect internal signal processing mechanisms.
- **Vertical vulnerability:** LLH Z continues to exhibit high susceptibility to orbit and timing-related fields. MT25's role in this metric, particularly the VC and residual terms, suggests these should be subject to focused validation in aviation-critical contexts.

This aligned interpretation allows for a better-informed prioritization of fuzzing campaigns and robustness assessments. Targeting high-impact fields like URA, UDREI, RTC, and orbit dynamics ensures efficient coverage of worst-case scenarios while avoiding unnecessary overhead from low-sensitivity parameters.

The revised matrix provides a consolidated, evidence-backed foundation for GNSS vulnerability testing, enabling

strategic selection of test inputs and parameter focus in line with observed system behavior under perturbation.

VIII. Conclusion and Future Work

This study presented a detailed sensitivity analysis of SBAS message parameters using a one-factor-at-a-time (OFAT) approach, aimed at improving GNSS receiver robustness through targeted perturbation and fuzz testing. The evaluation spanned key receiver outputs, including protection levels, dilution of precision, navigation state continuity, clock bias, signal gain control, and positional accuracy in both ECEF and LLH coordinate domains.

The findings reveal that certain SBAS parameters—particularly **MT9_URA**, **MT6_UDREI**, **MT10_RTC**, **MT9_ClockParam**, and **MT24_FC_F1**—exert high and consistent influence across multiple output metrics. These fields represent critical levers for both integrity performance and system vulnerability and should be prioritized in validation and fuzzing frameworks.

Additionally, the analysis uncovered strong coupling between horizontal positioning metrics (ECEF X/Y and LLH X/Y), which responded similarly to clock and correction-related perturbations. The vertical domain (LLH Z), by contrast, showed distinct and exaggerated sensitivity, especially to orbit velocity and code fields in **MT25**. This highlights the challenge of robust height estimation under SBAS augmentation and the need for dedicated testing in safety-of-life scenarios such as aircraft approaches.

The role of configuration-level fields, such as **MT12_Common_a0**, was also significant—particularly in influencing PDOP and long-term trajectory divergence—indicating the value of these less frequently targeted parameters in systematic robustness testing.

Future Work will extend this framework along several dimensions:

- **Multi-parameter analysis:** Incorporating global sensitivity techniques (e.g., Sobol indices or Morris method) to capture interaction effects and nonlinear parameter interdependencies.
- **Automated fuzzing pipelines:** Integrating data-driven or ML-based fuzzers guided by the sensitivity matrix to generate high-impact test cases efficiently.
- **System diversity:** Applying the methodology to different classes of GNSS receivers, including low-cost modules and high-integrity aviation systems, to explore generalizability.
- **Resilience modeling:** Augmenting fuzzing efforts with recovery time and fail-safe trigger analysis to assess not only system vulnerability but also robustness in failure response.

By linking empirical sensitivity findings with practical fuzzing design, this study contributes to the development of smarter, more efficient GNSS validation tools—offering a pathway to enhanced safety and resilience in increasingly complex navigation environments.

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